

Reverse genetics of sexual antagonism

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Individually silencing 125 fruit fly genes reveals opposing fitness effects of mutations between females and males, as well as between germline and somatic tissues.

Evolutionary adaptation is conventionally thought to happen via natural selection on beneficial genetic variants, but variants that are beneficial in some environments can be harmful in others. For example, a genetic variant of the *DsFAR2-B* gene encodes a pheromone that makes male fruit flies more attractive to females, yet this same molecule reduces the survival of females¹. In other words, the genetic variant is ‘sexually antagonistic’ because it cannot simultaneously accommodate the evolutionary interests of females and males². Writing in *Nature Ecology & Evolution*, Xia and colleagues³ find evidence for sexually antagonistic selection in the fruit fly *Drosophila melanogaster* after individually knocking down 125 genes and measuring their female and male fitness effects.

Over the past decade, researchers have taken diverse approaches to identifying sexually antagonistic genes. One approach uses genome-wide association studies, in which the frequencies of naturally occurring genetic variants are correlated with sex differences in fitness^{4,5}. Other approaches scan genomes for variant frequency differences between females and males^{6,7}, experimentally impose sex-differential selection and track ensuing frequency changes^{8,9}, or measure sex-specific fitness after natural selection has been removed altogether¹⁰.

Instead, Xia et al. apply a reverse genetic approach to study sexually antagonistic effects. They use a workhorse of *Drosophila* genetics: the GAL4–UAS system, in which nearly any gene can be silenced in nearly any tissue¹¹. Put simply, the system works by crossing flies that carry a GAL4 transcriptional activator protein (which can be expressed in a specific tissue) with flies that carry a UAS–RNAi construct that uses RNA interference to silence a specific gene (Fig. 1). The authors performed about 15,000 such crosses to silence the expression of 125 genes in either somatic or germline tissues. They then estimated fitness as the total number of adult offspring produced by knockdown

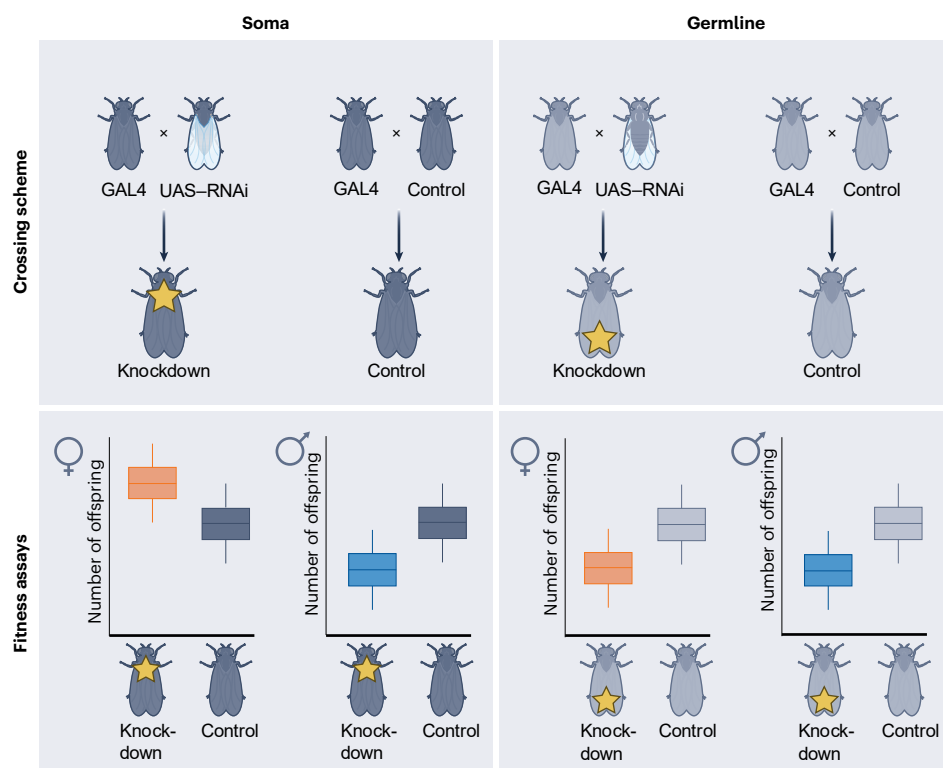


Fig. 1 | Reverse genetics of sexual antagonism. a, Schematic of the genetic crosses used by Xia et al. Fly lines that carry a GAL4 promoter were crossed to fly lines that carry a UAS–RNAi construct (or controls) to silence each of 125 genes in somatic (left) or germline (right) tissues. **b**, Modal outcome, averaged across

all genes, of female (orange) and male (blue) fitness assays, illustrated as box plots (62% of genes were sexually antagonistic in the soma; 41% were sexually concordant in the germline).

females or males, relative to controls. Their main finding is that silencing the expression of genes in somatic tissues typically reduces the fitness of males, but typically increases the fitness of females (Fig. 1). Of the 125 genes silenced, 62% showed a pattern of sexually antagonistic selection, in which fitness decreased in one sex and increased in the other. Among these sexually antagonistic mutations, 87% harmed males and benefitted females: a battle of the sexes indeed.

These results align well with evidence for sexually antagonistic fitness effects in specific genes, such as the fruit-fly gene duplicate *Apollo* and *Artemis*¹². However, the pervasiveness of the effect is puzzling, given previous evidence that spontaneous mutations are typically deleterious in both sexes¹⁰. One potential clue lies in comparing somatic and germline tissues (for example, ovaries and testes). In marked contrast to knockdowns of somatic tissues, Xia et al. found that 41% of genes were sexually concordant when silenced in germline tissues, and only 8% were sexually antagonistic (Fig. 1). Perhaps the benefits of expressing genes in the female germline outweigh the costs of expressing them in somatic tissues, and there is no sexual antagonism when considering total fitness. Or perhaps the soma–germline conflict coexists with sexually antagonistic selection on total fitness. Either way, conflict appears to have an important role in the function of many fruit fly genes.

As with any novel approach, the application of reverse genetics to study sexually antagonistic selection has pros and cons. Notably, gene knockdowns are an excellent tool if the goal is to reconstruct the null variant that existed before the evolution of each gene. Yet theories of sexually antagonistic variation were mostly developed with single-nucleotide variants in mind and not the large-effect mutations that Xia et al. generated, which leaves an important gap between their data and theory. This gap may contribute to the absence of an effect for two proposed correlates of sexual antagonism – genomic location (autosomal versus X linkage¹³) and degree of sex-biased gene expression¹⁴ – in Xia et al.'s data. Another limitation is their choice of genes. Among around 14,000 fruit fly genes, they mostly chose

young genes (<35 million years old), which have unusual functions, expression patterns and rates of evolution¹⁵. Although this choice is informed by practical considerations (for example, old genes are often essential and may render flies inviable when silenced) and has little apparent effect on sexual antagonism in their data, it limits generalizations. Finally, Xia et al. leave open the questions of what phenotypes mediate sexually antagonistic effects, how these effects are partitioned across fitness components (for example, survival versus reproduction), and how they combine across germline and somatic tissues. Future work will help to paint a clearer picture of the conflicts they have uncovered.

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Competing interests

The author declares no competing interests.